

Samiur R. Mir

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EDUCATION

Oklahoma State University

Ph.D. in Physics

Oklahoma, United States

Aug. 2021 – Present

University of Alberta

M.Sc. in Physics

Alberta, Canada

Sept. 2019 – Mar. 2021

University of Dhaka

B.Sc. in Physics

Dhaka, Bangladesh

Jan. 2015 – Aug. 2019

PUBLICATIONS & PREPRINTS

4. Vedran Brdar, Dibya S. Chattopadhyay, **Samiur R. Mir**, Tousif Raza, and Marc S. Romanowski, “Prospects for Neutrino Observation and Mass Measurement from Binary Neutron Star Mergers”, [arXiv:2511.16658 \[hep-ph\]](#) (under PRD review)
3. **Samiur R. Mir**, Carlos A. Argüelles, K.S. Babu, Vedran Brdar, “Double Bangs in IceCube as a Window to the Neutrino Mass Origin”, [arXiv:2511.07541 \[hep-ph\]](#) (under PRL review)
2. Vedran Brdar, Ying-Ying Li, **Samiur R. Mir**, Yi-Lin Wang, “Collider Prospects for the Neutrino Magnetic Moment Portal”, [JHEP 10 \(2025\) 230](#), [arXiv:2502.07024 \[hep-ph\]](#)
1. M. Jamil Aslam, Wen Chen, Andrzej Czarnecki, **Samiur R. Mir**, Muhammad Mubasher, “Rare Decays of the Positronium Ion and Molecule, $\text{Ps}^- \rightarrow e^- \gamma$ and $\text{Ps}_2 \rightarrow e^+ e^- \gamma, \gamma \gamma, e^+ e^-$ ”, [Phys. Rev. A 104, 052803 \(2021\)](#), [arXiv:2108.06785 \[hep-ph\]](#)

WORKS IN PROGRESS

1. Vedran Brdar, **Samiur R. Mir**, “Probing New Physics with Ultra-High-Energy Neutrinos”, anticipated time of completion: Jan 2026
2. Carlos A. Argüelles, Vedran Brdar, Dibya S. Chattopadhyay, **Samiur R. Mir**, “Neutrino–Dark Matter Interactions in Light of High-Energy Neutrino Data”, anticipated time of completion: April 2026
3. K.S. Babu, Vedran Brdar, **Samiur R. Mir**, “Search for Running PMNS Matrix at Colliders”, anticipated time of completion: April 2026

ACHIEVEMENTS

Student Travel Awards granted by APS (Texas, Arkansas, Oklahoma Section) Fall Meeting	2025
Student Travel Awards granted by APS (Division of Particles & Fields) and Pheno organizers	2024
Recipient of Honours Scholarship (top 5% CGPA in Bachelors) University of Dhaka	2019
National Olympiad Champion Bangladesh National Math Olympiad	2013

TECHNICAL SKILLS

Programming Languages: Python, C++

Computational Software: FORM, Mathematica

Packages: FeynRules, FeynCalc, MadGraph, SNOwGLOBES, nuSQuIDS

CONFERENCE PRESENTATIONS

Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube	2025
Particle Physics on the Plains, University of Kansas	
Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube	2025
APS Meeting Texas, Arkansas, Oklahoma Section, Texas Christian University	
Unveiling the Neutrino Magnetic Moment: Constraints from Colliders	2025
Pheno Symposium, University of Pittsburgh	
Unveiling the Neutrino Magnetic Moment: Constraints from Colliders	2025
LPPC seminar at Harvard University	
Neutrino Magnetic Moment at Colliders	2024
Particle Physics on the Plains, University of Kansas	
Collider Probes of Neutrino Magnetic Moment	2024
DPF - Pheno Symposium, University of Pittsburgh	
Single Photon Decay of the Positronium Ion	2020
57th Winter Nuclear and Particle Physics Conference, Banff, Alberta	
How to Find Decay Distribution from Symmetry	2019
10th Annual Symposium for Graduate Physics Research, University of Alberta	

SUMMER SCHOOLS AND PROFESSIONAL TRAININGS

International Neutrino Summer School <i>Fermilab</i>	Aug. 2025
NVIDIA Fundamentals of Deep Learning workshop <i>Oklahoma State University</i>	June 2023
Python Boot Camp <i>University of Alberta</i>	Oct. 2019
Computational Atomic Astrophysics <i>University of Dhaka</i>	Oct. 2017
Asian Science Camp <i>Universiti Tunku Abdul Rahman</i>	Aug. 2017

LEADERSHIP ACTIVITIES

Coordinator and Contributor, HEP-pheno Student Seminar	2023 – 2025
<i>Dept. of Physics, Oklahoma State University</i>	
Vice-President, Physics Graduate Student Association	2023 – 2025
<i>Dept. of Physics, Oklahoma State University</i>	
Outreach Volunteer, Physics Open House	2023 – Present
<i>Dept. of Physics, Oklahoma State University</i>	
Academic Team Member <i>Bangladesh Mathematical Olympiad</i>	2015 – 2019
Academic Instructor <i>Community of Physics, Bangladesh</i>	2015 – 2019

TEACHING EXPERIENCE

Oklahoma State University	
Teaching Assistant: Electricity & Magnetism (Labs & Discussions)	Aug. 2021 –Present
University of Alberta	
Teaching Assistant: Particles and Waves, Electricity and Magnetism (Labs)	Aug. 2019–April 2021

MENTORING EXPERIENCE

Marc Romanowski, The College of New Jersey
NSF REU Visiting Undergraduate Researcher

June 2025 –Aug. 2025

REFERENCES

- **Prof. Vedran Brdar** (Ph.D. Supervisor)
Assistant Professor
Department of Physics
Oklahoma State University
Email: vedran.brdar@okstate.edu
- **Prof. Carlos Argüelles-Delgado**
Assistant Professor
Department of Physics
Harvard University
Email: carguelles@g.harvard.edu
- **Prof. Kaladi S. Babu**
Regents Professor of Physics
Department of Physics
Oklahoma State University
Email: kaladi.babu@okstate.edu